

POKHARA UNIVERSITY

Level: Bachelor

Semester: Spring

Year: 2025

Programme: BCSIT

Full Marks: 100

Course: Data Structure and Algorithm

Pass Marks: 45

Time: 3 hrs.

Candidates are required to answer in their own words as far as practicable. The figures in the margin indicate full marks.

Section "A"

Very Short Answer Questions

Attempt all the questions. $[10 \times 2 = 20]$

1. Define ADT and list its types.
2. What is base case in recursion?
3. Define Stack as an Abstract Data Type (ADT). Mention its fundamental operations.
4. Define priority queue.
5. What are the advantages of linked list over an array?
6. Differentiate between internal and external sort.
7. Differentiate between internal and external sorting.
8. Define hash function with an example.
9. Differentiate between directed and undirected graphs. Provide a labeled diagram for each.
10. Define Big Oh notation and explain why Big Oh is Used?

Section "B"

Descriptive Answer Questions

Attempt any five questions. $[5 \times 10 = 50]$

11. Explain Tower of Hanoi. Write a recursive algorithm for solving a tower of Hanoi and explain its working along with example.
12. Write an application of stacks. Convert the following infix expression to postfix showing the content of the stack at each step.
 $A \wedge B * (C + (D * E) - F) / G$
13. Why do we need to balance the binary search here? Justify with an example. Create an AVL tree from the data: 24, 12, 8, 15, 35, 57, 40, 45, 78
14. Write an algorithm to create single linked list with example.
15. Define hashing, hash function, and hash table. Using the hash function $h(k) = k \% 11$ and a hash table of size 11, insert the following keys: 5, 40, 18, 22, 16, 30, 27, 49, 39.
Perform the insertion using:
 - a. Linear probing
 - b. Quadratic probing

16. Explain the concept of Divide and Conquer algorithm. Hand traces the Quick Sort algorithm for the array: 78, 34, 21, 43, 7, 18, 9, 56, 38, 19, 48, 53 also, mention its time complexity.

Group "C"

Long Answer Questions

Attempt any two questions. $[2 \times 15 = 30]$

17. Write a C program to perform the following operations on a Singly Linked List:
- Insert an element at the end of the list
 - Delete an element from the beginning of the list
 - Display all the elements of the list
18. Explain hashing. For given input {11, 26, 49, 16, 1, 64, 18, 0, 9, 15} and hash function $H(X) = X \bmod 10$, show the resulting hash table using:
- Linear Probing
 - Quadratic Probing
19. a. Write a program in C to create and traverse the nodes in linked list (assume number of nodes = 4)
- b. What is Huffman coding? Calculate:
- Huffman code of each character
 - Average code length of:
 - Length of Huffman encoded data
- For {A=45, B=13, C=12, D=16, E=9, F=5}